⌈Commutator

⌈Def. Let G be a group. Given $x, y \in G$, define

$$[x, y] \overset{\text{def}}{=} x^{-1}y^{-1}xy$$

is called the Commutator of x and y. And, given two subset A and B,

$$[A, B] \overset{\text{def}}{=} \langle [a, b] \mid a \in A,\ b \in B \rangle$$

Particularly, $G' \overset{\text{def}}{=} [G, G]$ is called the derived set.

⌈Prop. G' char G, (So $G' \trianglelefteq G$).

Proof. Given $\sigma \in \mathrm{Aut}(G)$ and $[x, y]$ for arbitrary $x, y \in G$,

$$\sigma([x, y]) = \sigma(x^{-1}y^{-1}xy) = \sigma(x)^{-1} \cdot \sigma(y)^{-1} \cdot \sigma(x) \cdot \sigma(y) = [\sigma(x), \sigma(y)]$$

So, $\sigma[[G, G]] = [G, G]$.

⌈Prop. Given $H \leq G$, $H \trianglelefteq G$ if and only if $[H, G] \leq H$

Proof. $H \trianglelefteq G \iff$ for all $g \in G$ and $h \in H$, $ghg^{-1} \in H$

$[H, G] \leq H \iff$ for all $g \in G$ and $h \in H$, $[h, g] = h^{-1}g^{-1}hg \in H$.

So proved.

⌈Prop. Given $N \trianglelefteq G$, G/N is abelian if and only if $G' \leq N$.

Proof. G/N is abelian iff for any $xN, yN \in G/N$, $xyN = yxN$

iff for any $x, y \in G$, $x^{-1}y^{-1}xy \in N$

iff $G' \leq N$.